

EPIGENETICS

Module map

- Module 10: Epigenetics
- Connected lab: lab-11_epigenetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 10

Epigenetics

1

Gene
regulation
as control

2

DNA
methylation

3

Histone
and
chromatin
change

4

Environment
and
expression

5

Epigenetics
versus
mutation

Lab evidence

lab-11_epigenetics.md

EPIGENETICS

Learning objectives

- Explain gene regulation as control of gene activity.
- Describe DNA methylation at an introductory level.
- Connect chromatin structure to gene accessibility.
- Give a careful example of environmental influence on expression.
- Distinguish epigenetic regulation from mutation.

OBJECTIVE 1

Explain gene regulation as control of gene activity.

OBJECTIVE 2

Describe DNA methylation at an introductory level.

OBJECTIVE 3

Connect chromatin structure to gene accessibility.

OBJECTIVE 4

Give a careful example of environmental influence on expression.

OBJECTIVE 5

Distinguish epigenetic regulation from mutation.

EPIGENETICS

Topic sequence

- Gene regulation as control: Cells regulate which genes are active rather than using every gene equally.
- DNA methylation: DNA methylation can reduce transcription without changing base sequence.
- Histone and chromatin change: Histone and chromatin changes alter access to DNA.
- Environment and expression: Environmental conditions can influence gene expression patterns.
- Epigenetics versus mutation: Epigenetic change affects gene activity, while mutation changes DNA sequence.

01

Gene regulation as control

Cells regulate which genes are active rather than using every gene equally.

02

DNA methylation

DNA methylation can reduce transcription without changing base sequence.

03

Histone and chromatin change

Histone and chromatin changes alter access to DNA.

04

Environment and expression

Environmental conditions can influence gene expression patterns.

05

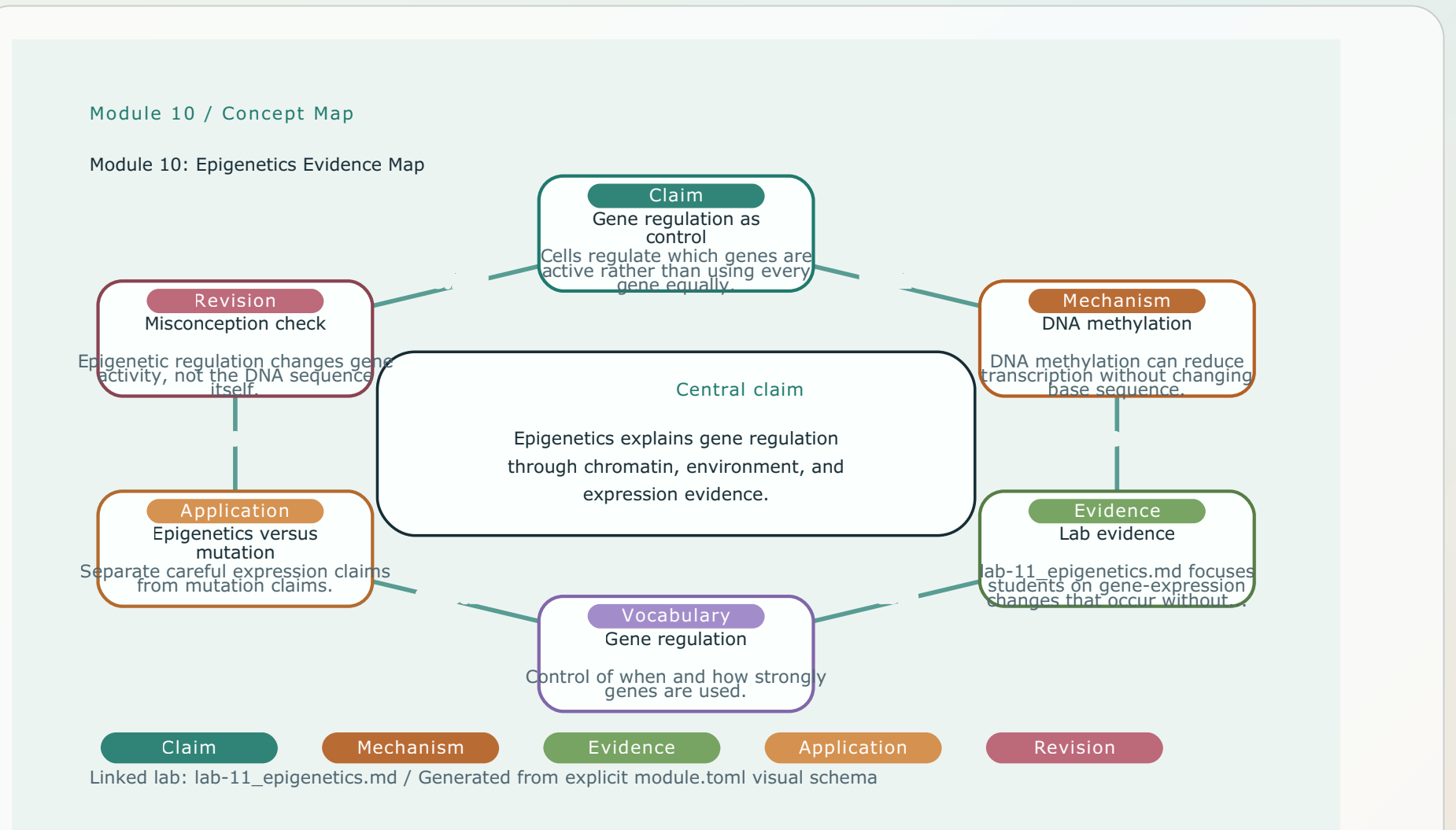
Epigenetics versus mutation

Epigenetic change affects gene activity, while mutation changes DNA sequence.

EPIGENETICS

Module 10: Epigenetics Evidence Map

- Module focus: Epigenetics
- Central claim: Epigenetics explains gene regulation through chromatin, environment, and expression evidence.
- Key nodes to connect: Gene regulation as control, DNA methylation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-11_epigenetics.md supplies an evidence surface for the map.



Teaching note: Ask students to explain one edge in their own words before moving on.

EPIGENETICS

Module 10: Epigenetics Reasoning Sequence

- Module focus: Epigenetics
- Trace the model: Gene regulation as control -> DNA methylation -> Histone and chromatin change -> Environment and expression
- Inputs become outputs: Gene regulation as control, DNA methylation -> Explain gene regulation as control of gene activity., Describe DNA methylation at an introductory level.
- Feedback check: lab-11_epigenetics.md evidence checks whether students can explain histone and chromatin change.
- Lab connection: lab-11_epigenetics.md gives students a process to observe or test.

Module 10 / Process Model

Module 10: Epigenetics Reasoning Sequence

Inputs

- Gene regulation as control
- DNA methylation
- Evidence from lab-11_epigenetics.md

Outputs

- Explain gene regulation as control of gene activity.
- Describe DNA methylation at an introductory level.
- Distinguish epigenetic regulation from mutation.

Reasoning sequence

1

Gene regulation as control

Cells regulate which genes are active rather than using the same gene equally.

2

DNA methylation

DNA methylation can regulate transcription without changing the base sequence.

3

Histone and chromatin change

Histone and chromatin changes alter access to DNA.

4

Environment and expression

Environmental conditions influence gene expression patterns.

5

Epigenetics versus mutation

Epigenetic change affects gene activity, while mutation changes the DNA sequence.

Feedback

lab-11_epigenetics.md evidence checks whether students can explain histone and chromatin change.; Retrieval questions reveal whether terms...

Constraint

Epigenetic regulation changes gene activity, not the DNA sequence itself.; Epigenetic change affects gene activity, while mutation changes DNA...

Linked lab: lab-11_epigenetics.md

EPIGENETICS

Terms as evidence handles

- Gene regulation: Control of when and how strongly genes are used.
- Epigenetics: Heritable or persistent expression changes without DNA sequence change.
- DNA methylation: Chemical marking of DNA often associated with reduced transcription.
- Histone: Protein around which DNA is packaged.
- Chromatin: DNA-protein complex that can be open or compact.
- Transcription factor: Protein that helps control transcription.

GENE REGULATION

Control of when and how strongly genes are used.

DNA METHYLATION

Chemical marking of DNA often associated with reduced transcription.

CHROMATIN

DNA-protein complex that can be open or compact.

EPIGENETICS

Heritable or persistent expression changes without DNA sequence change.

HISTONE

Protein around which DNA is packaged.

TRANSCRIPTION FACTOR

Protein that helps control transcription.

EPIGENETICS

Lab connection

- Lab file: lab-11_epigenetics.md
- Lab evidence should test or illustrate:
Histone and chromatin change
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Gene regulation as control

EVIDENCE

lab-11_epigenetics.md

CLAIM

Explain gene regulation as control of gene activity.

EPIGENETICS

Contrast check

- Do not stop at: Cells regulate which genes are active rather than using every gene equally.
- Stronger explanation: Epigenetic change affects gene activity, while mutation changes DNA sequence.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Cells regulate which genes are active rather than using every gene equally.

DEEPER

Epigenetic change affects gene activity, while mutation changes DNA sequence.

EPIGENETICS

Module 10: Epigenetics Retrieval and Lab Check

- Module focus: Epigenetics
- Retrieval prompt: Why do cells regulate genes?
- Answer check: Explain gene regulation as control of gene activity.
- Required terms: Gene regulation, Epigenetics, DNA methylation
- Lab connection: lab-11_epigenetics.md; lab-11_epigenetics.md supplies evidence for gene-expression changes that occur without changing DNA sequence.

Module 10 / Retrieval Card

Module 10: Epigenetics Retrieval and Lab

Check

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

Routine

Cover notes -> answer aloud
-> cite evidence -> revise

Why do cells regulate genes?

Check: Explain gene regulation as control of gene activity.

How can two cells with the same DNA act differently?

Check: Describe DNA methylation at an introductory level.

What is DNA methylation?

Check: Connect chromatin structure to gene accessibility.

How can chromatin structure affect transcription?

Check: Give a careful example of environmental influence on expression.

Terms to use

Gene regulation, Epigenetics, DNA methylation, Histone, Chromatin

Lab connection

lab-11_epigenetics.md supplies evidence for gene-expression changes that occur without changing DNA...

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

EPIGENETICS

Practice quiz bridge

- 1. How does epigenetic regulation differ from a mutation?
- 2. DNA methylation is usually associated with:
- 3. When chromatin is packed tightly, genes in that region are generally:
- 4. Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

QUESTION 1**Key: C**

How does epigenetic regulation differ from a mutation?

QUESTION 2**Key: A**

DNA methylation is usually associated with:

QUESTION 3**Key: D**

When chromatin is packed tightly, genes in that region are generally:

QUESTION 4**Key: B**

Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

EPIGENETICS

Synthesis and exit ticket

- Exit claim: explain gene regulation as control using evidence from lab-11_epigenetics.md.
- Must include: Gene regulation, Epigenetics, DNA methylation.
- Revision prompt: what evidence would change your explanation?

CLAIM

Gene regulation as control

EVIDENCE

lab-11_epigenetics.md

REVISION

What would change your mind?